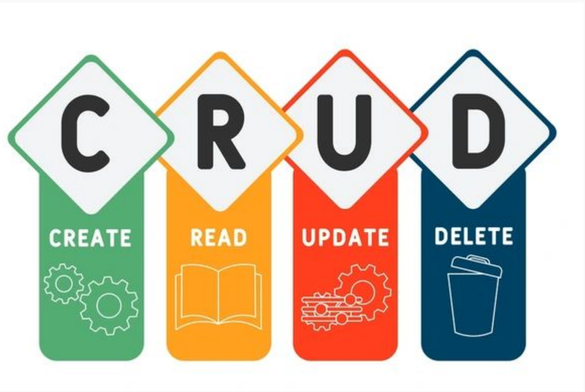
Google Developer Group
Universitas Sriwijaya

Learning 14

CRUD Operations & Mutations

Synchronizing UI and database operations



CRUD in Database Systems

Most applications require CRUD operations

CRUD stands for:

- Create → add new student
- Read → view students
- Update → edit student data
- Delete → remove student

Today we implement full CRUD.

Create Data in Supabase

Example: inserting a new student

```
const { data, error } = await supabase
  .from("students")
  .insert([
    {
      name: "Rina Putri",
      major: "Computer Science",
      year: 2024
    }
  ])
```

Equivalent SQL operation:

```
INSERT INTO students (name, major, year)
VALUES ('Rina Putri', 'Computer Science', 2024);
```

Update Data in Supabase

Example: inserting a new student

```
const { data, error } = await supabase
  .from("students")
  .update({ major: "Software Engineering" })
  .eq("id", 1)
```

Equivalent SQL operation:

```
UPDATE students
SET major = 'Software Engineering'
WHERE id = 1;
```

Delete Data in Supabase

Example: deleting a student.

```
const { error } = await supabase
  .from("students")
  .delete()
  .eq("id", 1)
```

Equivalent SQL:

```
DELETE FROM students
WHERE id = 1;
```

These operations modify server data.

Why Mutations Are Needed

In the previous session we used queries to read data.

```
useQuery()
```

But queries should not modify server data.

TanStack Query provides mutations for operations that change data

```
Queries → Read data
Mutations → Create / Update / Delete
```

Creating a Mutation

Example: creating a student.

```
const mutation = useMutation({
  mutationFn: createStudent
})
```

Trigger mutation:

```
mutation.mutate(newStudent)
```

TanStack Query handles:

- loading state
- succes state
- error state

Syncing UI with Database

After creating or updating data, the UI must update.

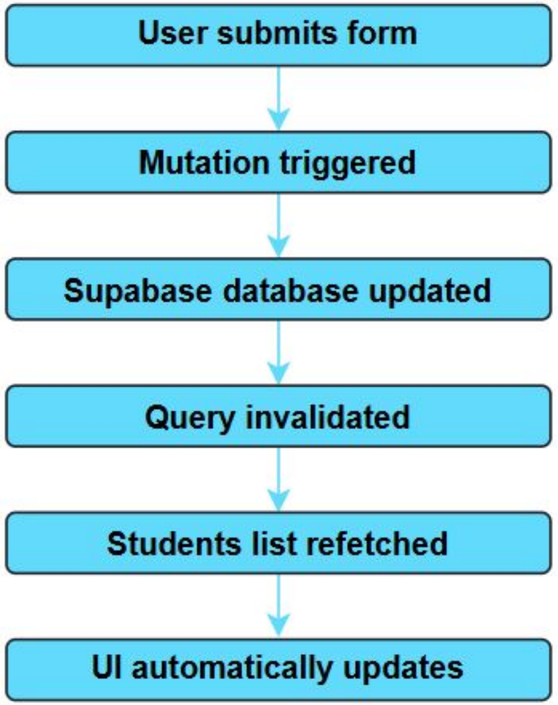
TanStack Query provides query invalidation.

```
queryClient.invalidateQueries({
  queryKey: ["students"]
})
```

This tells TanStack Query to refetch updated data.

Example Flow

After creating or updating data, the UI must update



No manual refetching required

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Thank You !